September 21–25, 2026

Professional Learning in Practice

Feedback-focused PLCs that move student learning forward

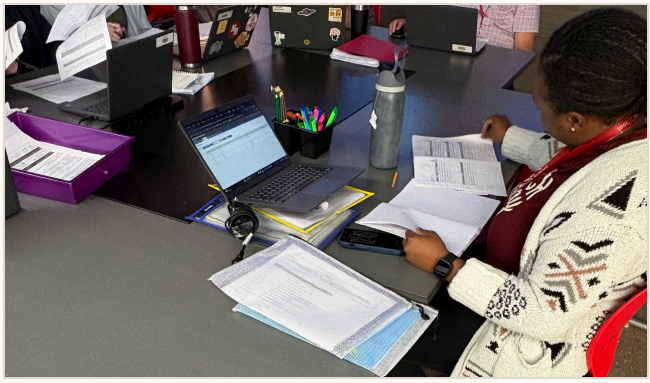
NEW BEDFORD HIGH SCHOOL · SCIENCE DEPARTMENT

SCIENCE DEPARTMENT PLC

From Student Work to Useful Feedback

This week's professional learning centered on a practical commitment: **student work is evidence.** Teams examined what students' work shows, identified a high-leverage next step, and planned feedback students can use to strengthen their thinking.

Rather than stopping at a score or a completed task, the feedback PLC asked teachers to consider what students understand now, what needs clarification, and how the next learning experience can make their thinking more visible.



Science teachers collaborate around student work to plan feedback and purposeful next steps.

THE FEEDBACK-FOCUSED PLC CYCLE

1 · ANALYZE
Look Closely at Student Work
Identify a learning pattern and reconnect it to the intended goal.

→

2 · RESPOND
Plan Feedback Students Can Use
Choose a clear next action and the support students need to take it.

→

3 · REPLAN
Strengthen the Next Experience
Make a focused revision that preserves rigorous scientific thinking.

Feedback that Prompts Action

Feedback matters because it helps students use evidence about their current work to improve what they do next. Research shows that feedback can be a powerful influence on learning, but its impact depends on the kind of information it provides and how students can act on it [1,2].

Effective feedback does more than identify a strength or need. It gives students a reachable next action: revise an explanation, correct a model, compare evidence, extend an idea, or apply learning in a new context.

- Keep feedback specific, actionable, and anchored in evidence.
- Protect student thinking—feedback should not complete the work for students.
- Name the new evidence that will show whether the next step helped.

The closing check: each team leaves with an evidence-based follow-up and a clear look-for that will show whether students moved forward.

FEEDBACK FOR OUR OWN GROWTH

A Culture of Continuous Improvement

Our department will also provide one another feedback with the shared goal of improving our teaching and leadership practices.

I welcome feedback wholeheartedly as an important part of my own personal growth, and I encourage all of us to offer feedback generously, respectfully, and with the intention of helping one another improve.

SOURCES & RESEARCH FINDING

Finding: Feedback is most useful when it connects goals, current progress, and next steps; a meta-analysis of 435 studies found a medium overall effect on student learning.

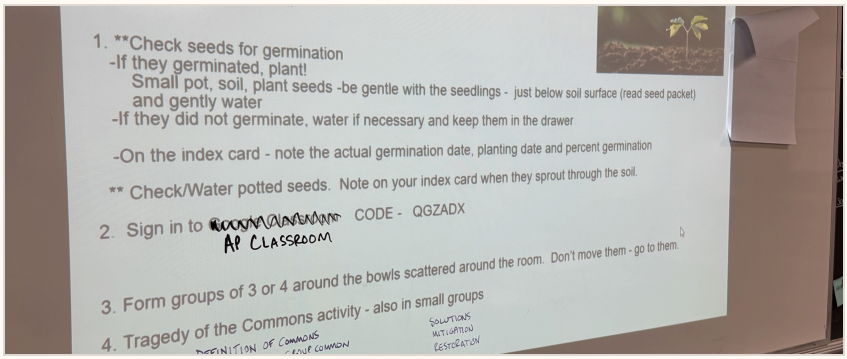
[1] Hattie & Timperley (2007), *Review of Educational Research, 77*(1), 81–112. doi:10.3102/003465430298487 [2] Wisniewski, Zierer, & Hattie (2020), *Frontiers in Psychology, 10*, 3087. doi:10.3389/fpsyg.2019.03087

PLC priority: read student work closely, provide feedback students can use, and replan the next learning experience.

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Classrooms & Curriculum in Practice

Inquiry, collaborative pacing, and standards-aligned materials



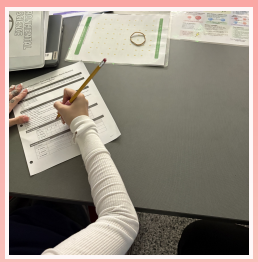
AP Environmental Science students launch the *Tragedy of the Commons* activity by examining a shared-resource dilemma.



AP Environmental Science students engage in discussion as they consider evidence, trade-offs, and shared-resource decisions.

FEATURED TOPICS

- AP Environmental Science explores shared resources and sustainability
- Environmental Science teachers co-lead curriculum development and pacing
- Biology OpenSciED teams protect rigorous, standards-aligned learning
- Student thinking remains the center of science instruction



Student work makes thinking visible for discussion and next steps.

AP ENVIRONMENTAL SCIENCE

Exploring a Shared Resource in Focus

In **Ms. Patricia Gifford's AP Environmental Science class**, students engaged in a discussion-centered *Tragedy of the Commons* activity that connected scientific reasoning to real community choices.

Students considered what can happen when many people depend on the same limited resource—from a fishery or forest to groundwater, clean air, or a shared public space. They weighed individual choices against collective outcomes, discussed trade-offs, and considered how data can help communities manage resources responsibly.

OBSERVE	EXPLAIN	DESIGN
Notice patterns in how a shared resource changes over time.	Use evidence to connect individual choices to group outcomes.	Propose fair, testable solutions that support sustainability.

Curriculum Collaboration in Practice

Ms. Cassidy Barrett, Ms. Lisa Richard, and their Environmental Science colleagues are co-leading curriculum development and pacing. Their shared work strengthens coherence across units while keeping investigations, evidence, and standards-aligned learning at the center.

Biology OpenSciED: Students are actively engaged while the team collaborates to ensure proper pacing and high-quality, standards-aligned materials. Shared planning protects rigorous, coherent science learning across the course.

Special thanks to Ms. Grace Stephenson for actively leading and reviewing the OpenSciEd curriculum to ensure standards alignment. She creates and shares additional resources with colleagues to plan for student learning and helps ensure that all OpenSciEd lessons remain aligned with standards.

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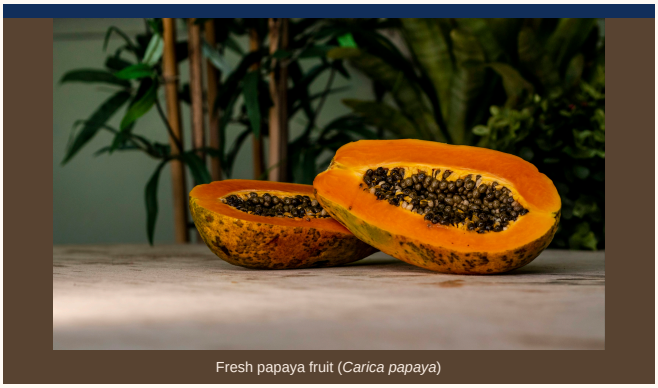
Science Corner

The biochemistry inside an everyday fruit

SCIENCE CORNER

Papaya's Protein-Splitting Toolkit

How plant proteases in Carica papaya turn peptide bonds into a powerful laboratory and food-science question.



Fresh papaya fruit (*Carica papaya*)

Not just "a digestive fruit"

Papaya is often associated with digestion, but the more precise question is: **Which molecules in the plant can cut proteins, where are they found, and how do they work?** The answer centers on **cysteine proteases**. Papain is best known, but papaya latex also contains chymopapain, caricain, and glycyl endopeptidase—enzymes that hydrolyze peptide bonds in proteins [1,2].

Proteins are chains of amino acids, and peptide bonds are their links. Proteases help water break those bonds into smaller fragments. Papain is a classic cysteine protease: biochemical work identified cysteine and histidine residues together at its active site, helping explain its reaction mechanism [3].

Why green papaya and latex matter

When a papaya plant is damaged, specialized laticifers release latex. Researchers identify papain, chymopapain, caricain, and glycyl endopeptidase as prominent latex enzymes [1]. Genome research also shows that these papain-like cysteine proteases form a distinctive group in *Carica papaya*, with many highly expressed in green fruit [2].

That is why unripe latex is often used as a source for purified papain in food and laboratory settings. **Eating ripe papaya is not the same experiment as working with a concentrated enzyme preparation.** Source tissue, extraction, and activity differ.

From protein chemistry to food science

Papain's broad protein-cleaving activity has food-processing uses, including meat tenderizing [4]. In class, students can model a protein as a linked chain, identify peptide bonds, and design a controlled investigation of enzyme-driven hydrolysis.

Nutrition snapshot: papaya in a smoothie

For a kitchen-scale reference, USDA nutrient lists show that 1 cup of raw papaya (1-inch pieces) provides **88.3 mg of vitamin C** and **54 µg of folate** [6]. The NIH lists recommended daily vitamin C amounts of 75 mg for adult women and 90 mg for adult men; needs vary by age, life stage, and smoking status [7].

> ***On a personal note, papain's protein-splitting biochemistry—and papaya's vitamin C and folate profile—are why I added papaya to my smoothies :)***

Evidence before health claims

"Digestive enzyme" is useful shorthand, not a cure-all claim. The body uses its own digestive enzymes, and papaya-latex products are evaluated separately from fresh fruit. A recent European Food Safety Authority assessment considers food-enzyme papain under specified conditions of use [5].

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September 21–25, 2026

Classroom Spotlight: Sea Lab

Clear routines, active participation, and feedback that moves learning forward

MODEL CLASSROOM VISIT

Focus Is Built—One Routine at a Time

During a recent visit to **Ms. Monica Shope's Sea Lab model classes**, students entered with purpose, started their work immediately, and sustained active engagement throughout the lesson. Clear expectations, purposeful instruction, and calm monitoring created an environment where every student knew what to do and why it mattered.

A Generous Share with the Team

Ms. Shope kindly shared a teaching-technique guide describing the practices that support student focus, participation, and accountability. The guide draws on *Teach Like a Champion* techniques adapted to the needs of her classroom.

Thank you, Ms. Shope, for opening your classroom and sharing practical strategies with the science team.

THREE CLASSROOM SYSTEMS THAT MADE LEARNING VISIBLE

01 · START WITH CLARITY

Purposeful Entry & Directions

Students entered, began a task right away, and received clear directions before moving into the activity. The routine protected learning time and made expectations transparent.

- **Do Now / Threshold:** a meaningful start from the first moment in the room
- **Classroom expectations:** visible, simple norms for participation and self-management
- **MVP and numbered steps:** clear expectations for movement, voice level, and the next action

02 · SUSTAIN PARTICIPATION

Active Monitoring & Student Voice

Students regularly processed their thinking with a partner, while Ms. Shope moved around the room, noticed participation, and redirected calmly when needed.

- **Turn-and-talk:** frequent structured opportunities to think aloud with a partner
- **Circulate and radar:** continual movement and whole-room awareness
- **100% and non-verbal redirection:** responsive routines that keep every learner with the task

03 · USE FEEDBACK AS A NEXT STEP

Check, Press, Improve

Feedback appeared during instruction—not only at the end. Students were asked to complete and clarify their answers, explain reasoning, and improve a response when needed.

- **Checks for understanding:** frequent evidence of both content learning and task understanding
- **Right Is Right and Stretch It:** precise answers followed by deeper thinking
- **Positive framing and strategic praise:** feedback that names what is working and what comes next

> ***"The most powerful routines do not replace student thinking—they make it possible for every student to enter the work, stay with it, and improve."***

ENTRY
Students begin with purpose and a clear first task.

ENGAGEMENT
Routines make discussion, participation, and accountability visible.

FEEDBACK
Specific next steps help students improve while staying in the thinking.

September 21–25, 2026

Science Team Shoutouts

Celebrating the colleagues who make our department stronger

A DEPARTMENT BUILT ON COLLABORATION

Everyday acts of preparation, generosity, mentorship, and shared problem-solving make a meaningful difference for our students and one another. This week, we celebrate the colleagues who are helping our science team thrive.

01 Materials Ready for Learning

Thank you to **Mr. Tyler Gilbert, Mr. Brian McCauly, Mr. Nathaniel Walker, and colleagues** who helped organize purchased materials, including notebooks, so every class—including OpenSciEd classrooms—is ready to go.

02 Sharing OpenSci Chemistry

Ms. Samantha Roche shared her OpenSci Chemistry lesson plans and resources and is supporting colleagues as they implement the curriculum for the first time.

03 Refining Chemistry Together

Ms. Margaret Henning is proactively collaborating with other chemistry teachers to refine her OpenSciEd and Pre-AP Chemistry lessons.

04 Mentorship in Action

Ms. Jayni Gryss is generously giving her time to serve as a mentor for Ms. Henning. Thank you for investing in a colleague's growth.

05 Leading the Genetics Work

Ms. Kaelyn Dahl took the initiative to refine the Genetics curriculum and its scope and sequence, strengthening the learning pathway for students.

06 Physics Learning, First-Year Leadership

Mr. Nicholas LeBlanc is proactively learning and analyzing the OpenSciEd Physics curriculum in his first year teaching it, with a positive impact on student learning and engagement—especially during Unit 1 electric-circuit analysis lessons.

07 Returning to Support Students

We appreciate **Ms. Christine Adams** for returning to support **Ms. Heather Bonner's** classes. Thank you for your time and generosity.

08 Building a Positive PLC Culture

Ms. Ava Gagnon and Ms. Patricia Gifford actively promote a positive PLC culture, helping colleagues stay on agenda and mindful of time allocation.

09 Intentional Co-Teaching

Mr. Kevin Flynn and Ms. Cykka Lovett continually discuss and adjust instructional practices so students in their co-taught Biology classes can learn effectively.

10 Sharing Digital Expertise

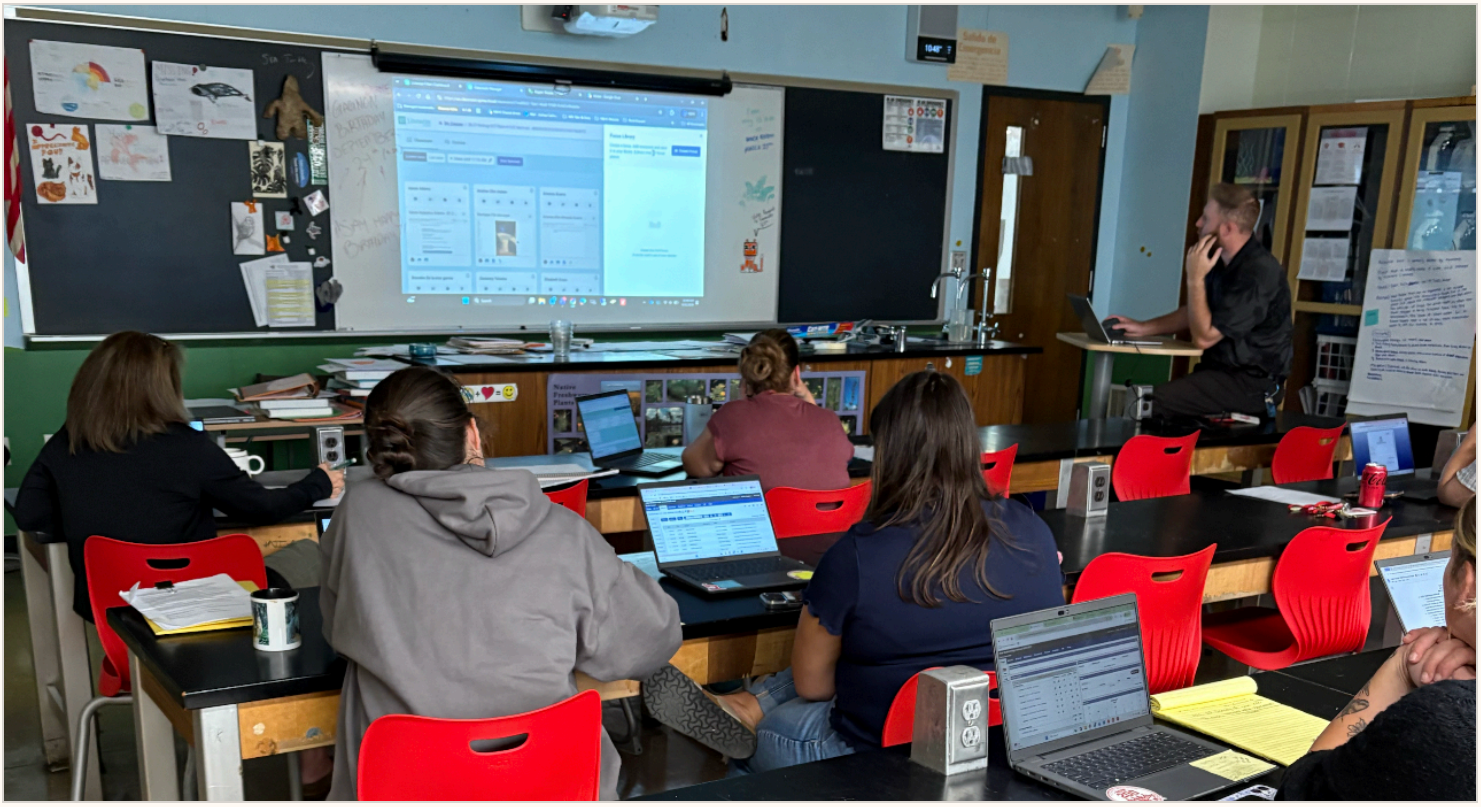
Mr. Jon Way continues to share his expertise in Google Classroom with colleagues, strengthening the tools that support learning every day.

And many more shoutouts than can fit on one page: these acts of teamwork, care, and generosity are a signature of the New Bedford High School Science Department. We are a team—and a family—that supports one another.

September 21–25, 2026

Professional Learning Shoutout

Linewize training to support engaged digital learning



Mr. Josh Carling facilitates a Linewize PD for the science team.

THANK YOU, MR. JOSH CARLING

Supporting Focused, Engaged Learning

Mr. Carling delivered a professional-development session to train science teachers to use Linewize and explore its classroom functions.

The session gave teachers an opportunity to consider how Linewize can support student engagement and focused learning during digital work.

A PRACTICAL TOOL FOR CLASSROOM ROUTINES

01 Explore

Learn the Linewize functions and how they connect to classroom routines.

02 Support

Use digital tools intentionally to promote focused student engagement.

03 Reflect

Consider how the tool can support productive, student-centered learning.

Our appreciation: thank you, Mr. Carling, for taking the time to share a practical resource with the New Bedford High School Science Department.

Professional learning is strongest when colleagues generously share tools, ideas, and strategies that help students thrive.